Harshita Sharma

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EDUCATION

B.Tech, Mechanical Engineering — IIT Bhilai | CGPA: **8.29**/10 | Best SGPA: **8.6** 2024 – **2028**

EXPERIENCE

Machine Learning Engineer Intern — APTransco, State Load Dispatch Centre *May – July 2026*

- **Improved on the centre's existing** forecaster, which failed when the SCADA feed went stale: replaced exact demand lags with rolling statistics and a staleness input so the model **degrades gracefully** — an **ambiguous, undefined problem** with no reference solution.
- **Designed and developed** a layered, **interface-driven Python** architecture with **swappable PostgreSQL/CSV backends**, so back-tested and deployed code paths stay identical.
- Contributed **automated data-quality checks and schema validation** over noisy multi-source feeds, removing silent failures and making runs reproducible.
- Implemented a **staleness-gated inverse-error weighting algorithm** over an XGBoost/LightGBM/Transformer ensemble with causal walk-forward recalibration, reaching **1.66%** MAPE next-block and 3.91% under a 24-hour telemetry blackout; **communicated trade-offs** to non-technical grid operators.

Tools: Python, PostgreSQL, SQL, XGBoost, LightGBM, PyTorch, Scikit-learn, Pandas, NumPy, Linux, Git

PROJECTS

Autonomous Precision Landing of Quadrotor — ROS 2 / PX4 Video | Repo

- Built a **multi-process ROS 2** (Humble) system on **Linux** where detector, controller, and world-setup nodes run as independent processes over DDS publish/subscribe **inter-process communication**, each debuggable in isolation.
- Implemented a **finite state machine** (INIT → TAKEOFF → LOITER_SEARCH → CENTERING_DESCENT → FINAL_LAND) driving the **PX4** autopilot through offboard control, with marker-loss recovery and a low-speed touchdown phase.
- Closed two independent **PID control loops** over pixel-space ArUco tracking error from an **OpenCV** detector; exposed all gains and thresholds as runtime parameters, not hard-coded constants.
- Performed **system-level integration**: PX4↔ROS 2 over the **uXRCE-DDS** (micro-ROS) bridge with vendored `px4_msgs`, camera-topic bridging, and marker spawning written against **both** Gazebo Classic and Ignition/Harmonic APIs.

Tools: Python, ROS 2 Humble, PX4 SITL, uXRCE-DDS, OpenCV, Gazebo, colcon, Linux, Git

Asteroid Risk Classification — Intra-IIT ML Challenge (1st Place) Repo

- Won **1st place** building an end-to-end risk-scoring pipeline on a heavily imbalanced dataset, from feature engineering through model evaluation.
- Traded overall accuracy (**82%**) against hazardous-class **recall** to match an asymmetric-cost decision.

Tools: Python, Scikit-learn, Random Forest, XGBoost, Pandas.

TECHNICAL SKILLS

Languages: Python, C++, C, Bash
CS Fundamentals: Data Structures, Object-Oriented Design, Complexity Analysis, Relational & Non-Relational Databases, Problem Solving
Software Engineering: Git/GitHub, Pull Requests & Code Review, Automated Validation & Testing, Debugging & Root-Cause Analysis, SDLC, Agile/Scrum, Linux
Distributed & Production Systems: Multi-process architectures, publish/subscribe messaging, service interfaces
ML & Data: PyTorch, Scikit-learn, XGBoost, LightGBM, CatBoost, Pandas, NumPy, Feature Engineering, Time-Series Forecasting

ACHIEVEMENTS AND LEADERSHIP

- **Maintainer, OpenLake (IIT Bhilai)** — own and help run an open-source project: review **pull requests** for code quality, triage issues, and coordinate distributed contributors via Git.
- **1st Place, Intra-IIT ML Challenge** [Repo] — end-to-end risk-scoring pipeline on a heavily imbalanced dataset, built in Python/Scikit-learn.
- **Gridlock Hackathon (Flipkart)** [Notebook] — stacked LightGBM/XGBoost/CatBoost ensemble forecasting city-wide demand; OOF R^2 **95.43** under leakage-free CV.
- **3rd Position, Summer ChainRxn Bootcamp** — developed an animated **front-end landing page** applying modern **UI/UX** principles [Repo 1] [Repo 2].
- **Competitive Programming — Data Structures & Algorithms in C++/C:** Codeforces (**120** problems) and LeetCode (**50 Days Badge 2026**; 75+ in **C++**)
- **Core Member**, Blockchain Society (IIT Bhilai). **MATLAB Certifications:** ML Onramp, Classification Models.